

GridPulse - AI-Assisted Grid Optimisation

Team: Renewabulls | Track: Sustainability | Lead: Khush Tailor

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Core Mission: Resolving renewable intermittency, protecting battery physical reserves, and eliminating operator cognitive overload in renewable microgrids.

The Microgrid Operator Dilemma

- Renewable Intermittency: Rapid generation swings threaten frequency stability (50.0 Hz).
- Inertia Constraints: 180 MW import limit cannot be breached without blackout risk.
- Battery Degradation: Discharging below 5% SOC (4 MWh) causes irreversible damage.
- Cognitive Overload: SCADA presents disjoint metrics across disparate screens.

Solution: Five Core Decision-Support Features

1. 72-Hour Demand Forecast: 288-step horizon at 15-min intervals \ (Seasonal Naive & Granite TTM\).
2. Renewable Performance Monitor: Real-time solar and wind PR% and curtailment tracking.
3. 12-Rule Anomaly Engine: Deterministic threshold classification \ (CRITICAL, WARNING, INFO\).
4. 72-Hour What-If Simulator: Physics-bounded balancing action evaluation \ (battery/load/import\).
5. AI Operator Brief: Plain-English synthesis answering six core operational questions.

System Architecture & Technology Integration

- AI Development & Interaction: IBM Bob / Antigravity assisted development \(\text{SKILL.md}\).
- Tool Interface Layer \(\text{MCP}\): Model Context Protocol server exposing 6 real-time tools.
- Application Runtime: Python FastAPI backend \(\text{165 automated tests passing}\).
- Optional Cloud AI: watsonx.ai Granite TTM and Instruct models \(\text{100\% offline fallbacks}\).

Physics Validation: True Energy & Feasibility Bounds

- Battery Physics Verification: 20 MW discharge for 1 hour from 50.2% SOC (40.16 MWh) yields exactly 25.2% SOC (20.16 MWh), maintaining 20.2% reserve above the 5% minimum floor.
- Feasibility Invariant: Infeasible requests (>180 MW import or >36.16 MWh battery drain) are strictly marked `is_feasible=false` with unmet power accounting.

Sustainability Impact & Production Roadmap

- Immediate Impact: Curtailment mitigation routes excess renewable energy to storage.
- Grid Resilience: Transforms reactive crisis firefighting into proactive 72-hour planning.
- Production Roadmap: Multi-bus AC power flow modeling & fleet aggregation via MCP tools.